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"Other-Play" for Zero-Shot Coordination

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Abstract

We consider the problem of zero-shot coordination - constructing AI agents that can coordinate with novel partners they have not seen before (e.g. humans). Standard Multi-Agent Reinforcement Learning (MARL) methods typically focus on the self-play (SP) setting where agents construct strategies by playing the game with themselves repeatedly. Unfortunately, applying SP naively to the zero-shot coordination problem can produce agents that establish highly specialized conventions that do not carry over to novel partners they have not been trained with. We introduce a novel learning algorithm called *other-play* (OP), that enhances self-play by looking for more robust strategies, exploiting the presence of known symmetries in the underlying problem. We characterize OP theoretically as well as experimentally. We study the cooperative card game Hanabi and show that OP agents achieve higher scores when paired with independently trained agents. In preliminary results we also show that our OP agents obtains higher average scores when paired with human players, compared to state-of-the-art SP agents.

1. Introduction

A central challenge for AI is constructing agents that can coordinate and cooperate with partners they have not seen before (Kleiman-Weiner et al., 2016; Lerer & Peysakhovich, 2017; Carroll et al., 2019; Shum et al., 2019). This is particularly important in applications such as cooperative game playing, communication, or autonomous driving (Foerster et al., 2016; Lazaridou et al., 2016; Sukhbaatar et al., 2016; Resnick et al., 2018). In this paper we consider the question of zero-shot coordination where agents are placed into a cooperative situation with a novel partner and must quickly coordinate if they wish to earn high payoffs.

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Our setting is a partially observed cooperative Markov game (MG) which is commonly known among both agents. The agents are able to construct strategies separately in the training phase but cannot coordinate on the strategies that they construct. They must then play these strategies when paired together one time. We refer to this as zero-shot coordination.

A popular way of constructing strategies for MG with unknown opponents is self-play (or "self-training"), (Tesauro, 1994). Here the agent controls both players during training and iteratively improves both players' strategies. The agent then uses this strategy at test time. If it converges, self-play finds a Nash equilibrium of the game and yields superhuman AI agents in two-player zero-sum games such as Chess, Go and Poker (Campbell et al., 2002; Silver et al., 2017; Brown & Sandholm, 2018). However, in complex environments self-play agents typically construct 'inhuman' strategies (Carroll et al., 2019). This may be a benefit for zero-sum games, but is less useful when it is important to coordinate with, not trick, one's partner.

Our main contribution is "other-play" (OP), an algorithm for constructing good strategies for the zero-shot coordination setting. We assume that with every MG we are provided with a set of symmetries, i.e. arbitrary relabelings of the state/action space that leave trajectories unchanged up to relabeling. One source of miscoordination in zero-shot settings is that agents have no good way to break the symmetries (e.g. should we drive on the left or the right?). In most MDPs, there are classes of strategies that require more or less coordinated symmetry breaking. OP's goal is to find a strategy that is *maximally robust* to partners breaking symmetries in different ways while still playing in the same class. OP works as follows: it uses RL to maximize reward when matched with agents playing the same policy under a random relabeling of states and actions under the known symmetries.

To show the intuition behind OP consider the following game: you need to coordinate with an unknown stranger by independently choosing one from a set of 10 different levers (Figure 1a). If both of you pick the same lever a reward of 1 point is paid out, otherwise you leave the game empty-handed. Clearly, without any prior coordination the only option is to pick one of the levers at random, leading to an expected reward of $1/10 = 0.1$.

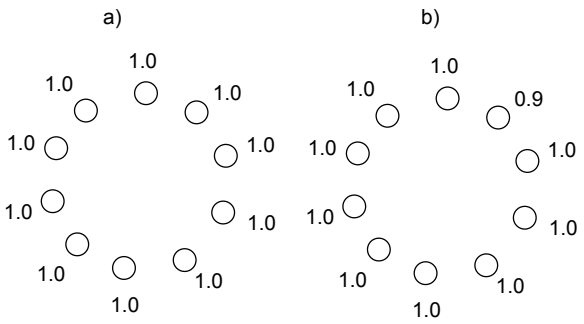

Figure 1. The *lever coordination game* illustrates the counter intuitive outcome of zero-shot coordination.

Next we consider a game that instead only pays 0.9 for one of the levers, keeping all other levers unchanged (Figure 1b). How does this change the coordination problem?

From the point of view of the MG the 1 payoff levers have no labels and so are symmetric. Since agents cannot coordinate on how to break symmetries, picking one of the 1.0 levers leads to 0.11 expected return. By contrast, OP suggests the choice of the 0.9 lever.

We note that this example illustrates another facet of OP: it is an equilibrium in meta-strategies. That is, neither agent wishes to deviate from OP as a reasoning strategy if the other agent is using it.

Note that OP does not use any action labels. Instead OP uses only features of the problem description to coordinate. Furthermore, note that the OP policy in this setting is the only policy that would *never* be chosen by the types of algorithms that try to use self-play to optimize team performance, e.g. VDN (Sunehag et al., 2018) or SAD (Hu & Foerster, 2019).

The main contributions of this work are: 1) we introduce OP as a way of solving the zero-shot coordination problem, 2) we show that OP is the highest payoff meta-equilibrium for the zero-shot coordination problem, 3) we show how to implement OP using deep reinforcement learning (deep RL) based methods, and 4) we evaluate OP in the cooperative card game Hanabi (Bard et al., 2020).

2. Related Work

2.1. Self-Play in Cooperative Settings

There is a large body of research on constructing agents to do well in positive-sum games. Self-play, if it converges, converges to an equilibrium of the game and so in purely cooperative games SP agents will be able to coordinate. Here the main problem is that SP may reach inefficient equilibria and so there is a large literature on pushing self-play toward higher payoff equilibria using various algorithmic innovations (Babes et al., 2008; Devlin et al., 2011; Devlin & Kudenko, 2016; Peysakhovich & Lerer, 2018). However, the setting where agents play with the same agents they have been trained with (aka. centralized training with decentralized execution) is quite different from the zero-shot coordination one which we study.

2.2. Cooperation and Coordination

A closely related problem to zero-shot coordination is ad-hoc teamwork (Stone et al., 2010; Barrett et al., 2011). For example: a robot agent joining another existing group of agents to play soccer (Barrett et al., 2011). Ad-hoc teamwork differs from the zero-shot coordination problem in that it is typically framed as a learning problem of learning the policies/capabilities of other agents during interaction whereas the pure zero-shot coordination scenario is one where there is no time to update a fixed policy that is constructed during training. These problems are closely linked and incorporating ideas from this literature into algorithms like OP is an interesting question for future research. However, another difference is that zero-shot agents only need to coordinate well with teams of agents that are optimized for the zero-shot setting, rather than arbitrary teams self-play of agents.

There is recent work looking at the situation where the one RL agent, trained separately, must join a group of new AI agents or humans (Lerer & Peysakhovich, 2018; Tucker et al., 2020; Carroll et al., 2019). These papers focus on using small amounts of observed behavior from partnered test time agents to either guide self-play to selecting the equilibrium (or "social convention") of the existing agents (Lerer & Peysakhovich, 2018; Tucker et al., 2020) or allow building a human model which can be used to learn an approximate best response using RL (Carroll et al., 2019). This setting is related, but zero-shot coordination gives no behavioral data to either agent to guide self-play or allow building a model of the other agent. Instead, zero-shot makes the assumption that test-time agents being themselves are optimized for the zero-shot setting (rather than the SP setting).

2.3. Game Theory and Tacit Coordination

Within behavorial game theory a large body of work considers coordination based on "focal points" or other shared grounding such as the famous "you lost your friend in New York City, where are you going to meet?" coordination problem (Schelling, 1980; Mehta et al., 1994). However, such focal points typically come from the fact that these coordination problems are not just abstract but grounded in *exogenous* features, *action labels*, that are meaningful due to a prior shared context. The zero-shot coordination set-

ting thus is a special form of the tacit coordination problem in which there are no shared exogenous features between the different agents and OP can be thought of as a way to coordinate in this setting.

There is also a large theoretical literature on learning and evolving coordination (Nowak, 2006). However, as with the self-play literature, it focuses on long run outcomes within a single group of agents learning or evolving together and does not typically focus on the question of engineering agents as we do.

2.4. Predicting Human Decision Making

Clearly, if we were able to accurately predict how our human counterparts are going to act in any given situation, the zero-shot coordination with human counterparts would reduce to learning a best response to those predicted actions. There is a large body of work using formal models to predict and understand human decision making (Camerer, 2011) and recent work that incorporates machine learning into this question (Wright & Leyton-Brown, 2010; Hartford et al., 2016; Peysakhovich & Naecker, 2017; Kleinberg et al., 2017; Fudenberg & Liang, 2019). However, the majority of this research focuses on extremely simple settings such as small normal form games (Wright & Leyton-Brown, 2010; Hartford et al., 2016; Fudenberg & Liang, 2019) or single decision problems (Peysakhovich & Naecker, 2017; Kleinberg et al., 2017) rather than complex cooperative settings with partial observability.

2.5. Domain Randomization

Our work is also related to the idea of domain randomization (Tobin et al., 2017). In RL and supervised learning domain randomization tries to make the realized model invariant to some feature of the environment. For example, an object detector should be invariant to the exact camera angle from which a view of an object is captured. OP applies a similar idea: a policy should be invariant to how an agent's partner breaks symmetries in the underlying game.

2.6. Exploiting Symmetries in Other Contexts

In the single agent context, it is harder to plan in MDPs that have more states. The idea of abstraction is to use underlying symmetries to 'compress' a large MDP into a simpler one, solve for the optimal strategy in the abstraction, and then lift the strategy to the original MDP. One set of such methods are MDP homomorphisms (van der Pol et al., 2020; Ravindran & Barto, 2004). These, like OP, use underlying symmetries but their goal is different: they want to find payoff maximizing policies for single agent decision problems, while OP seeks to find robust policies for zero-shot coordination. Note that as the lever game illustrates robust policies are not necessarily the payoff maximizing ones. In addition, these methods do not solve the problem of equilibrium selection among 'symmetric' policies in games, because the symmetry in the MDP just becomes a symmetry in the homomorphism.

A similar technique (compress, solve, then lift) is also used for finding Nash equilibria in large games like poker (Gilpin & Sandholm, 2007). In this case the abstraction treats 'isomorphic' states equally and thus reduces the effective number of states in the game. Again, the goal is different - poker abstractions are trying to find Nash equilibrium strategies in the original game while OP uses symmetries to select among a set of possible equilibria.

3. Zero-Shot Coordination

In this paper we study fully cooperative Markov games. To construct this environment we start out with a Dec-POMDP (Nair et al., 2003) with states $s_t \in \mathcal{S}$. There are $i = 1, \cdots N$ agents who each choose actions, $a_t^i \in \mathcal{A}$ at each time step.

The game is partially observable, $o_t^i \sim O(o|i, s_t)$ being each agent's stochastic observation function. At time t each agent has an action-observation history $\tau_t^i = \{o_0^i, a_0^i, r_0^i, \cdots, o_t^i\}$ and selects action a_i^t using stochastic policies of the form $\pi_\theta^i(a^i|\tau_t^i)$. The transition function, $P(s'|s, \mathbf{a})$, conditions on the joint action, $\mathbf{a}$.

The game is fully cooperative, agents share the reward r_t which is conditioned on the joint action and the state. Thus, the goal is to maximize the expected return $J = \mathbb{E}_\tau R(\tau)$, where $R(\tau) = \sum_t \gamma^t r_t$ is calculated using the discount factor γ.

Most work on cooperative MARL focuses on a setting where agents are trained together, although they must execute their policies independently at least at test time , *e.g.* (Lowe et al., 2017; Foerster et al., 2018a;b). The goal is to construct *learning rules*, i.e. functions that map Markov games to (joint) policies that select policies for each agent that together maximize expected discounted return. Because agents are trained together, these policies may be arbitrarily complex.

We are instead interested in achieving high returns with partners that were not trained together with our agent. Instead, we will frame the problem as follows: suppose that multiple independent AI designers will construct agents that have to interact in various but ex-ante unknown Dec-POMDPs without being able to coordinate beforehand, what learning rule should these designers agree on? To make this even more concrete, consider the case of independent autonomous vehicles made by multiple firms which have to interact in novel traffic situations on a daily basis.